

[RI/2024/Q2]

2 Name your Jupyter Notebook as:

TASK2_<your name>_<class>_<index number>.ipynb

For each of the sub-tasks, add a comment statement at the beginning of the code using the hash symbol '#' to indicate the sub-task the program code belongs to, for example:

```
In [1]: #Task 2.1
        Program code
```

Output:

Task 2.1

Write an insertion sort function that takes three parameters: `data_list`, `index`, `ascending` and returns a sorted `data_list`.

- `data_list` is a list of tuples.
- `index` is an integer to specify an index within a tuple.
- `ascending` is a Boolean flag.

The function will sort `data_list` based on the values at the specified `index` within the tuples. The `ascending` parameter determines the order of the sort: `True` for ascending order, `False` for descending order.

The function specification is:

```
FUNCTION insertion_sort(data_list: List[Tuple], index: Integer,
ascending: Boolean) RETURN List[Tuple]
```

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Example of `insertion_sort` function call:

```
data_list = [(1, 'Alice', 85), (2, 'Bob', 90), (3, 'Charlie', 80)]
sorted_list = insertion_sort(data_list, 2, ascending=False)
print(sorted_list)
# Output: [(2, 'Bob', 90), (1, 'Alice', 85), (3, 'Charlie', 80)]
```

Task 2.2

At an international coding competition, the results of the participants were recorded in a text file `CODING_COMPETITION.txt`.

Each record in the file has the following format, with each field separated by a comma: `<participant_id>,<participant_name>,<country>,<score>`

Write a function `task2_2` that takes a `filename` as parameter, reads the comma-separated data from the file and returns the data as a list of tuples.

Example: After calling function `task2_2`, these sample records in the file:

```
101,Jong Tan,Singapore,210
102,Mei Ling,Singapore,190
```

will be returned as

```
[('101','Jong Tan','Singapore',210),('102','Mei Ling','Singapore',190)]
```

Test your function by calling `task2_2("CODING_COMPETITION.txt")`.

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[RI/2024/Q2 Cont ']

Task 2.3

Write a function `task2_3` that performs the following:

- Use the list of tuples returned from the function `task2_2`.
- Sort the list by country in alphabetical order, using the `insertion_sort` function.
- Within each country, sort the participants by their scores in descending order using the `insertion_sort` function.
- Write the sorted data to a text file `SCORE_BY_COUNTRY.txt` in the format:
`<country>,<participant id>,<participant name>,<score>`.
- Each record should be on a new line.

The output file should contain lines of record that is first sorted by country, then by score. Sample lines in output file:

```
Australia,249,James Scott,235
Australia,267,Liam Taylor,230
Australia,139,Daniel Miller,225
.....
Singapore,101,Jong Tan,210
.....
```

Run your function `task2_3`.

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Save your Jupyter Notebook for Task 2.